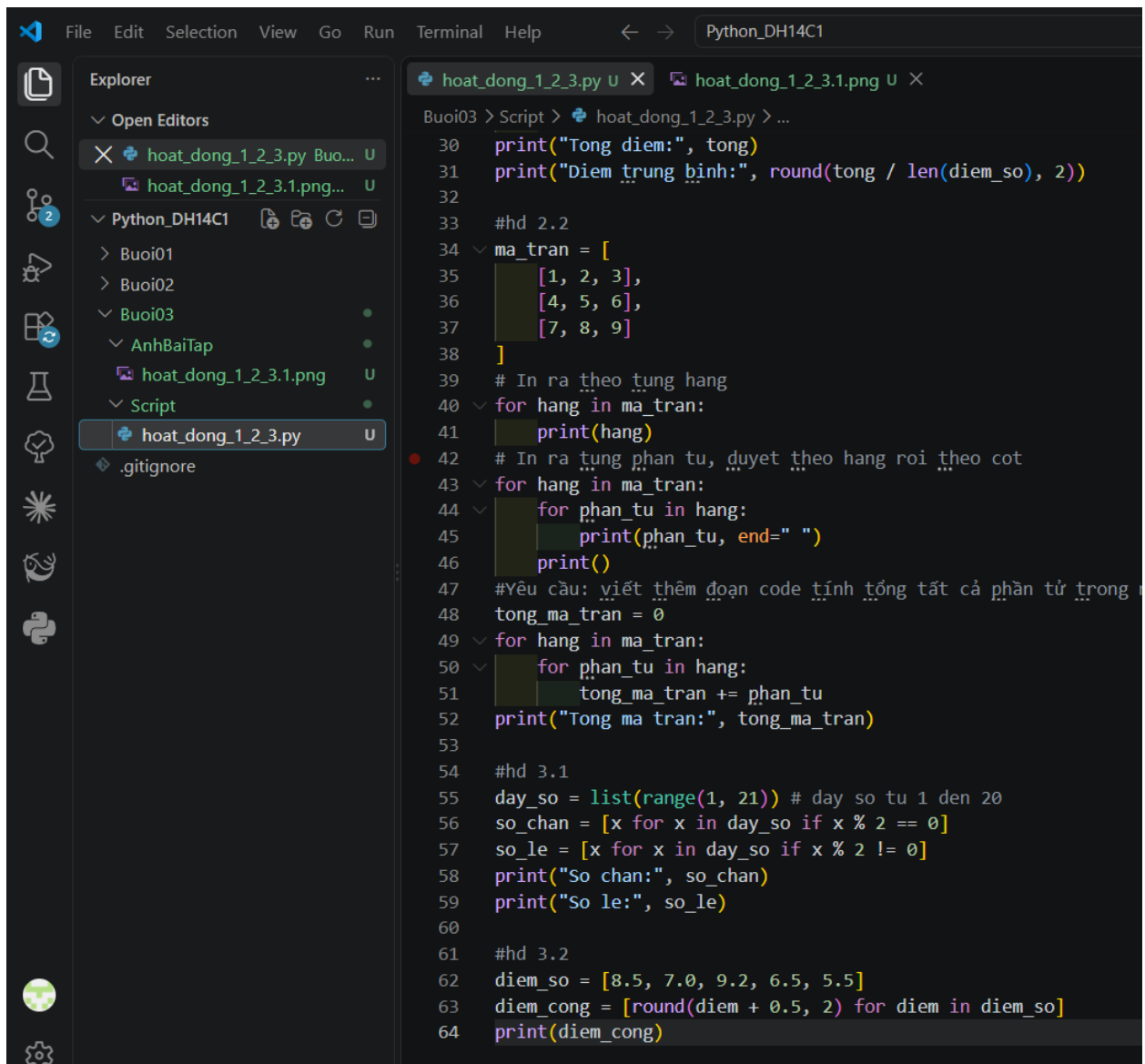


BÁO CÁO

câu hỏi hoạt động 1:

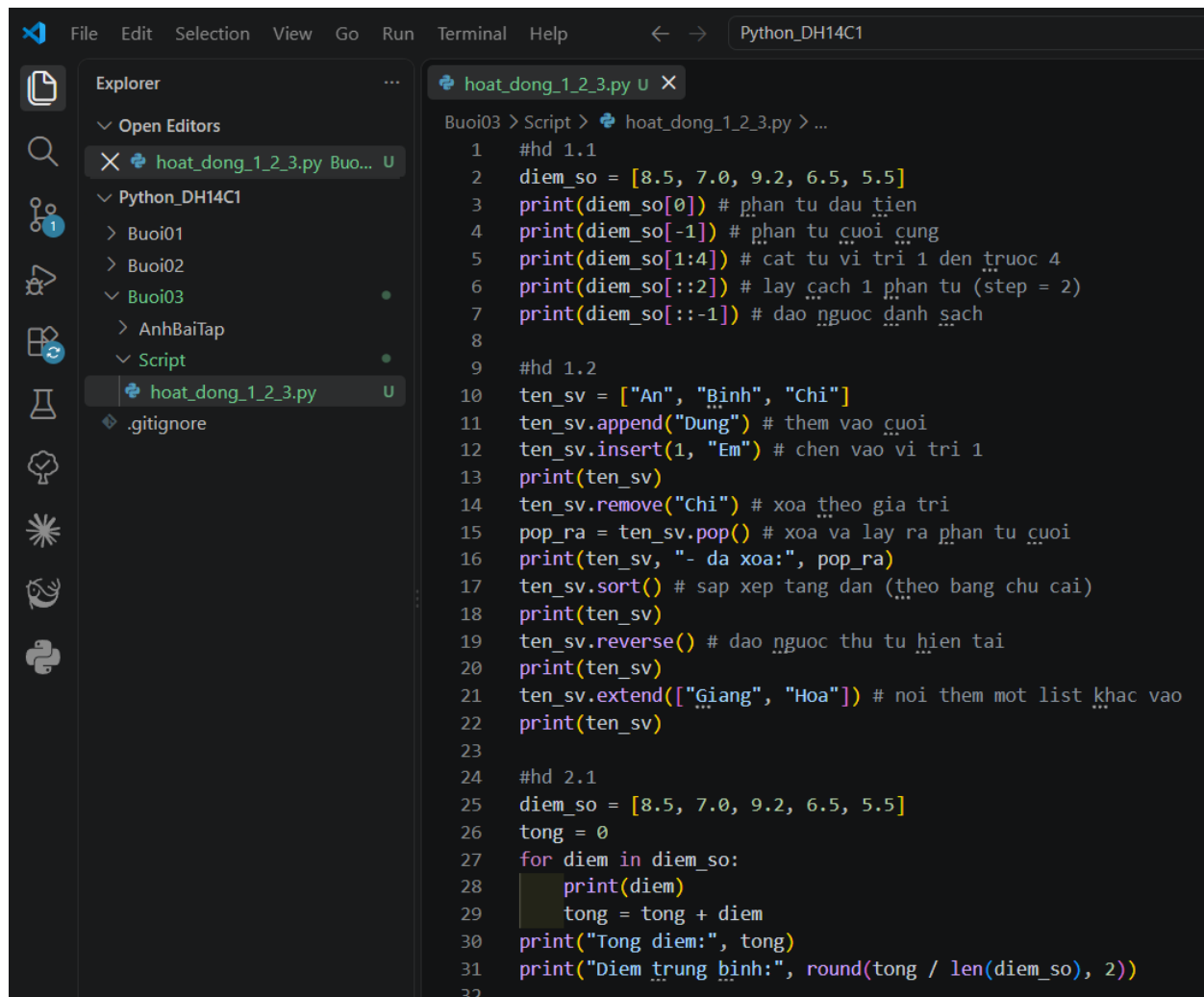
Sự khác nhau cốt lõi giữa `remove()` và `pop()`:

- `remove(x)`: Xóa theo giá trị `x` (tìm phần tử bằng `x` đầu tiên để xóa), không trả về giá trị vừa xóa.
- `pop(i)`: Xóa theo vị trí `i` (mặc định xóa phần tử cuối cùng), có trả về giá trị vừa xóa để lưu vào biến khác.



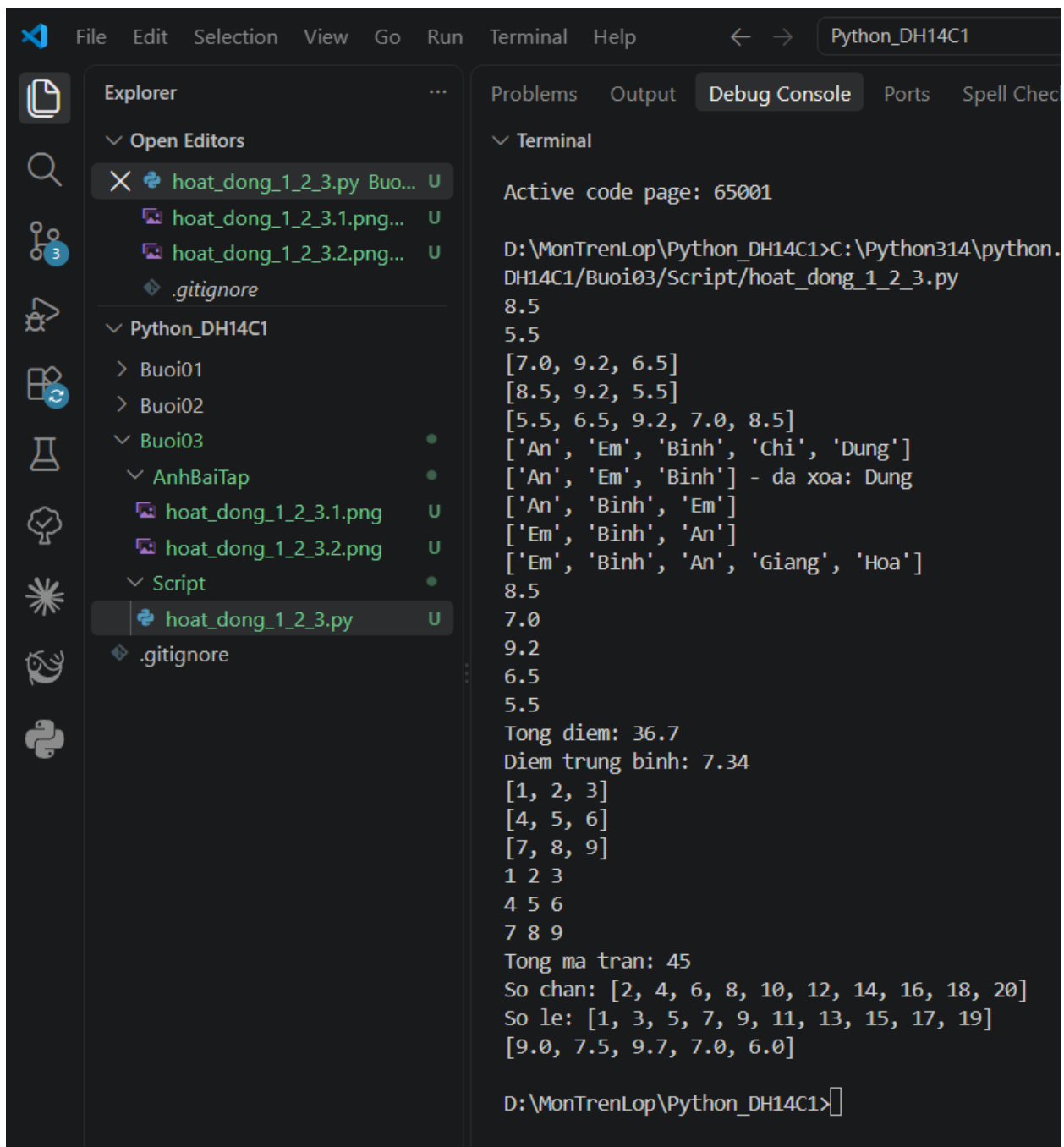
The image shows a Visual Studio Code editor window with a dark theme. The Explorer sidebar on the left displays a project structure for 'Python_DH14C1'. It includes folders 'Buoi01', 'Buoi02', and 'Buoi03'. Under 'Buoi03', there are files 'AnhBaiTap', 'hoat_dong_1_2_3.1.png', and a 'Script' folder. The 'Script' folder is expanded, showing 'hoat_dong_1_2_3.py' as the active file. The main editor area shows the content of 'hoat_dong_1_2_3.py', which is a Python script. The script includes comments in Vietnamese and performs several tasks: it prints the sum and average of a list, iterates through a 2D array, calculates the sum of elements, and filters even and odd numbers from a range. The script is currently at line 64.

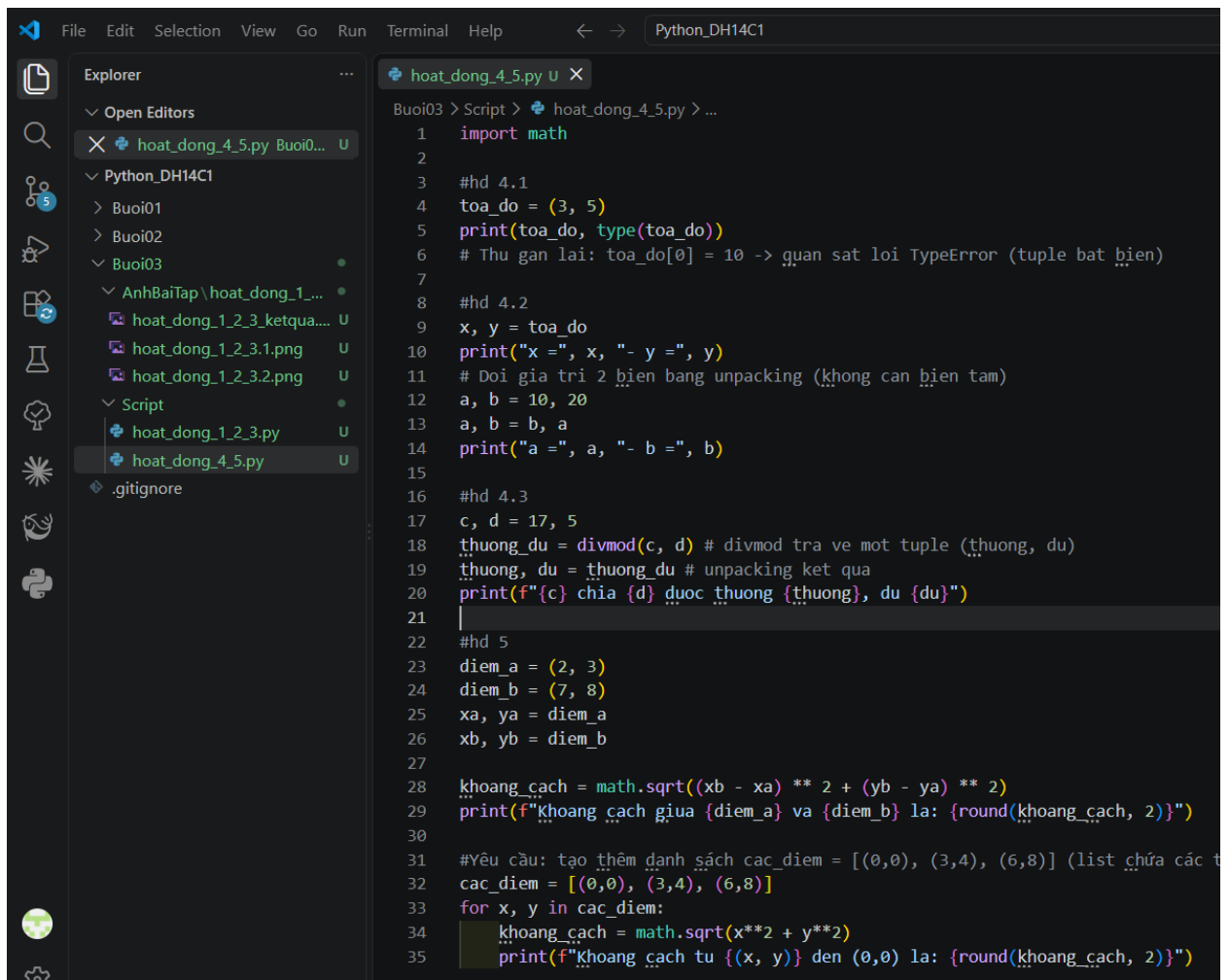
```
30 print("Tong diem:", tong)
31 print("Diem trung binh:", round(tong / len(diem_so), 2))
32
33 #hd 2.2
34 ma_tran = [
35     [1, 2, 3],
36     [4, 5, 6],
37     [7, 8, 9]
38 ]
39 # In ra theo tung hang
40 for hang in ma_tran:
41     print(hang)
42 # In ra tung phan tu, duyet theo hang roi theo cot
43 for hang in ma_tran:
44     for phan_tu in hang:
45         print(phan_tu, end=" ")
46     print()
47 #Yêu cầu: viết thêm đoạn code tính tổng tất cả phần tử trong
48 tong_ma_tran = 0
49 for hang in ma_tran:
50     for phan_tu in hang:
51         tong_ma_tran += phan_tu
52 print("Tong ma tran:", tong_ma_tran)
53
54 #hd 3.1
55 day_so = list(range(1, 21)) # day so tu 1 den 20
56 so_chan = [x for x in day_so if x % 2 == 0]
57 so_le = [x for x in day_so if x % 2 != 0]
58 print("So chan:", so_chan)
59 print("So le:", so_le)
60
61 #hd 3.2
62 diem_so = [8.5, 7.0, 9.2, 6.5, 5.5]
63 diem_cong = [round(diem + 0.5, 2) for diem in diem_so]
64 print(diem_cong)
```



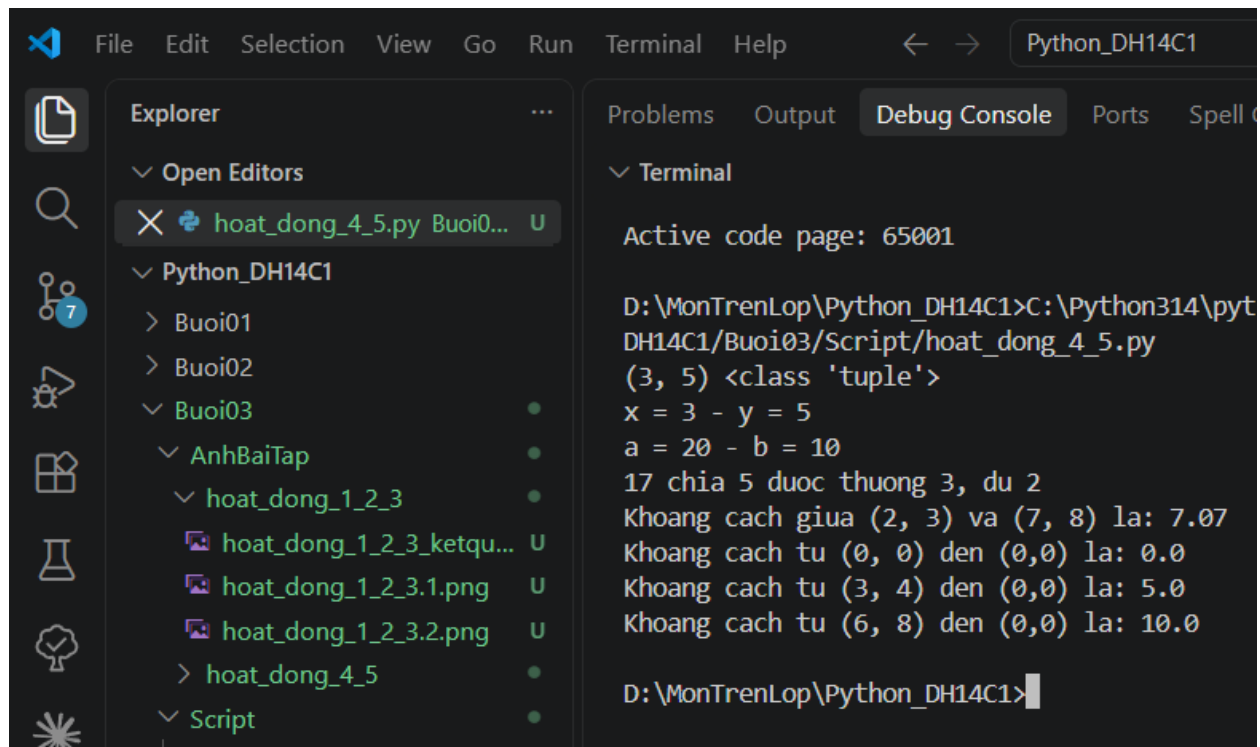
The image shows a Visual Studio Code editor window with a dark theme. The Explorer sidebar on the left displays the project structure for 'Python_DH14C1', including folders 'Buoi01', 'Buoi02', 'Buoi03', and 'AnhBaiTap', and a 'Script' folder containing the file 'hoat_dong_1_2_3.py'. The main editor area shows the code for 'hoat_dong_1_2_3.py' with line numbers 1 through 32. The code includes comments in Vietnamese and performs various list operations on 'diem_so' and 'ten_sv'.

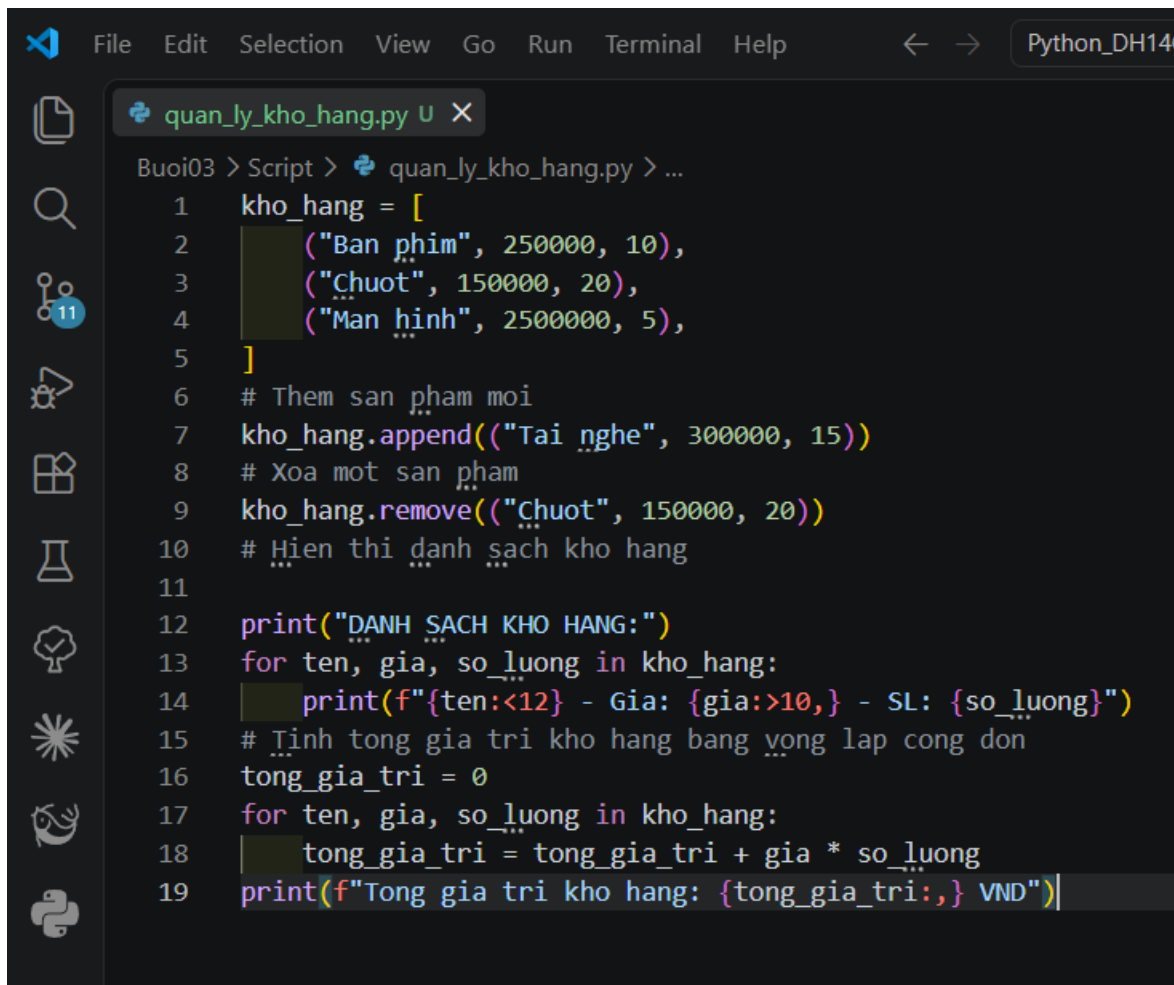
```
1  #hd 1.1
2  diem_so = [8.5, 7.0, 9.2, 6.5, 5.5]
3  print(diem_so[0]) # phần tử đầu tiên
4  print(diem_so[-1]) # phần tử cuối cùng
5  print(diem_so[1:4]) # cắt từ vị trí 1 đến trước 4
6  print(diem_so[::2]) # lấy cách 1 phần tử (step = 2)
7  print(diem_so[::-1]) # đảo ngược danh sách
8
9  #hd 1.2
10 ten_sv = ["An", "Bình", "Chi"]
11 ten_sv.append("Dung") # thêm vào cuối
12 ten_sv.insert(1, "Em") # chèn vào vị trí 1
13 print(ten_sv)
14 ten_sv.remove("Chi") # xóa theo giá trị
15 pop_ra = ten_sv.pop() # xóa và lấy ra phần tử cuối
16 print(ten_sv, "- đã xóa:", pop_ra)
17 ten_sv.sort() # sắp xếp tăng dần (theo bảng chữ cái)
18 print(ten_sv)
19 ten_sv.reverse() # đảo ngược thứ tự hiện tại
20 print(ten_sv)
21 ten_sv.extend(["Giang", "Hoa"]) # nối thêm một list khác vào
22 print(ten_sv)
23
24 #hd 2.1
25 diem_so = [8.5, 7.0, 9.2, 6.5, 5.5]
26 tong = 0
27 for diem in diem_so:
28     print(diem)
29     tong = tong + diem
30 print("Tong diem:", tong)
31 print("Diem trung binh:", round(tong / len(diem_so), 2))
32
```





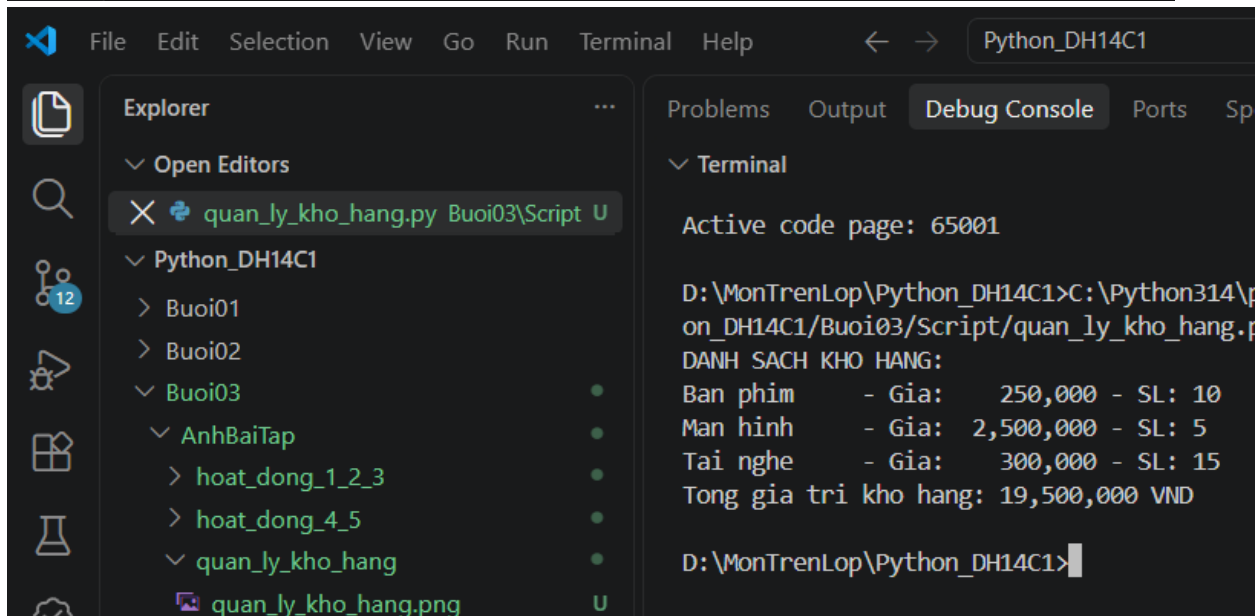
```
1 import math
2
3 #hd 4.1
4 toa_do = (3, 5)
5 print(toa_do, type(toa_do))
6 # Thu gan lai: toa_do[0] = 10 -> quan sat loi TypeError (tuple bat bien)
7
8 #hd 4.2
9 x, y = toa_do
10 print("x =", x, "- y =", y)
11 # Doi gia tri 2 bien bang unpacking (khong can bien tam)
12 a, b = 10, 20
13 a, b = b, a
14 print("a =", a, "- b =", b)
15
16 #hd 4.3
17 c, d = 17, 5
18 thuong_du = divmod(c, d) # divmod tra ve mot tuple (thuong, du)
19 thuong, du = thuong_du # unpacking ket qua
20 print(f"{c} chia {d} duoc thuong {thuong}, du {du}")
21
22 #hd 5
23 diem_a = (2, 3)
24 diem_b = (7, 8)
25 xa, ya = diem_a
26 xb, yb = diem_b
27
28 khoang_cach = math.sqrt((xb - xa) ** 2 + (yb - ya) ** 2)
29 print(f"Khoang cach giua {diem_a} va {diem_b} la: {round(khoang_cach, 2)}")
30
31 #Yeu cau: tao them danh sach cac_diem = [(0,0), (3,4), (6,8)] (list chua cac t
32 cac_diem = [(0,0), (3,4), (6,8)]
33 for x, y in cac_diem:
34     khoang_cach = math.sqrt(x**2 + y**2)
35     print(f"Khoang cach tu {(x, y)} den (0,0) la: {round(khoang_cach, 2)}")
```





```
File Edit Selection View Go Run Terminal Help Python_DH14C1

quan_ly_kho_hang.py U X
Buoi03 > Script > quan_ly_kho_hang.py > ...
1 kho_hang = [
2     ("Ban phim", 250000, 10),
3     ("Chuot", 150000, 20),
4     ("Man hinh", 2500000, 5),
5 ]
6 # Them san pham moi
7 kho_hang.append(("Tai nghe", 300000, 15))
8 # Xoa mot san pham
9 kho_hang.remove(("Chuot", 150000, 20))
10 # Hien thi danh sach kho hang
11
12 print("DANH SACH KHO HANG:")
13 for ten, gia, so_luong in kho_hang:
14     print(f"{ten:<12} - Gia: {gia:>10,} - SL: {so_luong}")
15 # Tinh tong gia tri kho hang bang vong lap cong don
16 tong_gia_tri = 0
17 for ten, gia, so_luong in kho_hang:
18     tong_gia_tri = tong_gia_tri + gia * so_luong
19 print(f"Tong gia tri kho hang: {tong_gia_tri:,} VND")
```



File Edit Selection View Go Run Terminal Help Python_DH14C1

Explorer

- Open Editors
 - quan_ly_kho_hang.py Buoi03\Script U
- Python_DH14C1
 - Buoi01
 - Buoi02
 - Buoi03
 - AnhBaiTap
 - hoat_dong_1_2_3
 - hoat_dong_4_5
 - quan_ly_kho_hang
 - quan_ly_kho_hang.png U

Problems Output Debug Console Ports Sp

Terminal

Active code page: 65001

```
D:\MonTrenLop\Python_DH14C1>C:\Python314\python.exe D:\MonTrenLop\Python_DH14C1\Buoi03\Script\quan_ly_kho_hang.py
DANH SACH KHO HANG:
Ban phim      - Gia:    250,000 - SL: 10
Man hinh      - Gia:   2,500,000 - SL: 5
Tai nghe      - Gia:    300,000 - SL: 15
Tong gia tri kho hang: 19,500,000 VND

D:\MonTrenLop\Python_DH14C1>
```

The screenshot shows the VS Code editor with the file explorer on the left and the editor window on the right. The file explorer shows a project named 'Python_DH14C1' with subfolders 'Buoi01', 'Buoi02', 'Buoi03', and 'AnhBaiTap'. The 'Buoi03' folder is expanded, showing files 'hoat_dong_1...', 'hoat_dong_4...', and 'Script'. The 'Script' folder is expanded, showing files 'hoat_dong_1...', 'hoat_dong_4...', and 'quan_ly_sinh_vien.py'. The 'quan_ly_sinh_vien.py' file is open in the editor window. The code in the editor is as follows:

```
Buoi03 > Script > quan_ly_sinh_vien.py > ...
1  danh_sach_sv = [(8.5, "An"), (7.0, "Binh"), (9.2, "Chi"), (6.5, "Dung")]
2  # Them sinh vien moi
3  danh_sach_sv.append((8.0, "Em"))
4  # Xoa mot sinh vien (biet chinh xac ca diem va ten)
5  danh_sach_sv.remove((7.0, "Binh"))
6  # Sua diem cho sinh vien o vi tri xac dinh (vi du vi tri 0)
7  danh_sach_sv[0] = (9.0, danh_sach_sv[0][1])
8  # Kiem tra mot sinh vien co trong danh sach hay khong (dung toan tu in)
9  print("Chi co trong danh sach khong?", (9.2, "Chi") in danh_sach_sv)
10 # Sap xep theo diem tang dan (mac dinh so sanh phan tu dau tien cua tuple truoc)
11 danh_sach_sv.sort()
12 print("Danh sach sau khi sap xep theo diem tang dan:")
13 for diem, ten in danh_sach_sv:
14     print(f"{ten} - {diem}")
15 # Sap xep giam dan
16 danh_sach_sv.sort(reverse=True)
17 print("Danh sach sau khi sap xep theo diem giam dan:")
18 for diem, ten in danh_sach_sv:
19     print(f"{ten} - {diem}")
```

The screenshot shows the VS Code editor with the file explorer on the left and the editor window on the right. The file explorer shows a project named 'Python_DH14C1' with subfolders 'Buoi01', 'Buoi02', 'Buoi03', and 'AnhBaiTap'. The 'Buoi03' folder is expanded, showing files 'hoat_dong_1...', 'hoat_dong_4...', and 'quan_ly_sinh_vien.py'. The 'quan_ly_sinh_vien.py' file is open in the editor window. The code in the editor is as follows:

```
Buoi03 > Script > quan_ly_sinh_vien.py > ...
1  danh_sach_sv = [(8.5, "An"), (7.0, "Binh"), (9.2, "Chi"), (6.5, "Dung")]
2  # Them sinh vien moi
3  danh_sach_sv.append((8.0, "Em"))
4  # Xoa mot sinh vien (biet chinh xac ca diem va ten)
5  danh_sach_sv.remove((7.0, "Binh"))
6  # Sua diem cho sinh vien o vi tri xac dinh (vi du vi tri 0)
7  danh_sach_sv[0] = (9.0, danh_sach_sv[0][1])
8  # Kiem tra mot sinh vien co trong danh sach hay khong (dung toan tu in)
9  print("Chi co trong danh sach khong?", (9.2, "Chi") in danh_sach_sv)
10 # Sap xep theo diem tang dan (mac dinh so sanh phan tu dau tien cua tuple truoc)
11 danh_sach_sv.sort()
12 print("Danh sach sau khi sap xep theo diem tang dan:")
13 for diem, ten in danh_sach_sv:
14     print(f"{ten} - {diem}")
15 # Sap xep giam dan
16 danh_sach_sv.sort(reverse=True)
17 print("Danh sach sau khi sap xep theo diem giam dan:")
18 for diem, ten in danh_sach_sv:
19     print(f"{ten} - {diem}")
```

The Debug Console shows the output of the script:

```
Active code page: 65001
D:\MonTrenLop\Python_DH14C1>C:\Python314\python.exe C1/Buoi03/Script/quan_ly_sinh_vien.py
Chi co trong danh sach khong? True
Danh sach sau khi sap xep theo diem tang dan:
Dung - 6.5
Em - 8.0
An - 9.0
Chi - 9.2
Danh sach sau khi sap xep theo diem giam dan:
Chi - 9.2
An - 9.0
Em - 8.0
Dung - 6.5
D:\MonTrenLop\Python_DH14C1>
```